

Yanze Wang

Academy of Mathematics and Systems Science, University of Chinese Academy of Sciences Ph.D.

Candidate in Pure Mathematics — expected June 2027

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Research & Practice

- **Mathematics:** birational geometry and the minimal model program — Sarkisov program, algebraically integrable foliations, moduli spaces.
- **AI practice:** local small projects with PyTorch / libtorch; agent-framework building.
- **AI4Math:** human verification of AI-generated mathematical results; research harnesses and Lean 4 formalization.
- **Engineering:** torch (Python & C++); Rust, C, LaTeX; familiar with Linux and Git; personal CLI / TUI programs developed with vibe coding (Rust).

Papers & Talks

Publications

Sarkisov Program for Algebraically Integrable Adjoint Foliated Structures.

Yifei Chen, Jihao Liu, and Yanze Wang. *International Mathematics Research Notices*, 2026(6): rnag045. 2026

Flop between algebraically integrable foliations on potentially KLT varieties.

Yifei Chen, Jihao Liu, and Yanze Wang. *International Journal of Mathematics*, 36(11):2550035. 2025

A Note on the Sarkisov Program.

Yifei Chen and Yanze Wang. *Higher Dimensional Algebraic Geometry: A Volume in Honor of V. V. Shokurov, LMS Lecture Note Series*, pp. 231–263, Cambridge University Press. 2025

Preprints

A klt generalized pair with infinitely generated canonical ring (AI-generated, human-verified).

Jihao Liu and Yanze Wang. *arXiv:2608.03258*. 2026

Twelve common flex lines in a general pencil of cubics (AI-generated, human-verified).

Jihao Liu and Yanze Wang. *arXiv:2607.26396*. 2026

A counterexample to the odd-dimensional rank bound for abelian p-group actions (AI-generated, human-verified).

Jihao Liu and Yanze Wang. *arXiv:2607.04891*. 2026

Invited Talks

Speaker, **Sarkisov Program for Algebraically Integrable and Threefold Foliations**, Xi'an Jiaotong- 2026
Liverpool University.

Projects

Programming

suckless — personal fork of dwm / dwmblocks (C), patched and maintained by hand, no AI assistance.

Calman — terminal task & event manager (CLI + TUI, Rust). CalDAV-compatible; sync delegated to git / vdirsyncer / rclone.

Markerss — TUI RSS reader (Rust, ratatui); Markdown export, URL-only bookmarks.

Lichtung — Hugo theme (Go templates / SCSS), powers two personal sites.

mumble — message board (JavaScript, Cloudflare Worker), embeddable.

AI Practice

Model training: local practice — pytorch-cpp, llm-from-scratch, build-llm-from-zero, tiny-reasoning-rl-lab, etc.

AiKit — agent framework: agents / skills / plugins for coding agents (pi, opencode, codex, claude code, ...).

Open-source: forks, improvements & usage — [Rethlas](#) (natural-language math reasoning), co-mathematician, etc.; contribution to the Lean 4 platform prove2.me.

Closed beta: agentero, math research workstation & paper reader (temporarily closed-source).

Education

Ph.D.

Mathematics PhD (in progress), AMSS, University of Chinese Academy of Sciences — expected June 2027.

Advisor: Prof. Yifei Chen.

Research: birational geometry — Sarkisov program, algebraically integrable foliations.

M.S.

Mathematics M.S., AMSS, University of Chinese Academy of Sciences.

Advisor: Prof. Yifei Chen.

Thesis: “Sarkisov program for foliated pairs”.

B.S.

Mathematics B.S., Beihang University.

Thesis: “Moduli space of curves”.

Undergraduate teaching assistant: Linear Algebra, Algebraic Geometry.

- MCM/ICM — Meritorious Winner (twice)
- Hua Luogeng Scholarship, Beihang University (twice)